RACE Regular Analysis Through Continuous Exercise

Atomic Structure: RACE-01

Discovery of subatomic particles (Discharge Tube Experiment; Cathode rays) Charge to mass ratio(e/m)

1. The specific charge of a proton is 9.6 x 10^7C kg^{-1}, then for an α-particle it will be-

(A) 2.4 × 10^7C kg^{-1} (B) 4.8 × 10^7C kg^{-1}

(C) 19.2 × 10^7C kg^{-1} (D) 38.4 × 10^7C kg^{-1}

CAS365

2. The correct order of specific charge ratio is :

(A) $Li^+ > Na^+ > F^-$ (B) $Na^+ > Mg^{2+} > Al^{3+}$

(C) $Mg^{2+} > F^- > Na^+$ (D) $F^- > Na^+ > Mg^{2+}$

CAS366

Atomic models (Thomson Model of Atom , Rutherford Nuclear Model of Atom)

3. A proton and a deuteron are projected towards the stationary gold nucleus, in different experiments, with the same speed. The distance of closest approach will be

(A) same (B) greater for proton

(C) greater for deuteron (D) depends on speed

CAS367

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4. Two particles, A and B, having same e/m ratio are projected towards silver nucleus, in different experiment, with the same speed. The distance of closest approach will be.

(A) Same for both (B) greater for A

(C) greater for B (D) depends on speed

CAS368

5. Suppose that an alpha particle of 3.20 MeV approaches head on a lead nucleus (Z = 82). Assuming that the lead nucleus remains at rest and the alpha particle momentarily comes to rest and reverses its direction at a distance much more than the radius of the lead nucleus, the distance of its closest approach is :

(A) 24.6 fm (B) 36.9 fm (C) 59.2 fm (D) 73.8 fm

CAS369

6. α -particles are projected towards the nucleus of following metals, with the same kinetic energy. Towards which metal, the distance of closest approach will be minimum?

(A) Cu (Z = 29) (B) Ag (Z = 47)

(C) Au (Z = 79) (D) Ca (Z = 20)

CAS370

Atomic Number and Mass Number

7. X^+, Y^{2+} and Z^- are isoelectronic with CO_2. Increasing order of protons in X^+, Y^{2+} and Z^- is :

(A) $X^+ = Y^{2+} = Z^-$ (B) $X^+ < Y^{2+} < Z^-$

(C) $Z^- < X^+ < Y^{2+}$ (D) $Y^{2+} < X^+ < Z^-$

CAS371

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8. It is known that an atom contains protons, neutrons and a electrons. If the mass of neutron is assumed to half of its original value where as that of proton is assumed to be twice of its original value then the atomic mass of $_7N^{14}$ will be :-

(A) same (B) 25 % more (C) 14.28 % more (D) 28.5 % less

CAS372

Developments leading to the Bohr's model of atom; Maxwell's Electromagnetic Wave Theory; Wave characteristics such as wavelength, frequency, time period, wave number; Electromagnetic Spectrum

9. The frequency of light for 1.0 Å wavelength is:-

(A) 3×10^{18} Hz (B) 3×10^{18} sec^{-1}

(C) 3×10^{18} Cycles per sec (D) All Of the above

CAS373

Particle Nature of Electromagnetic Radiation: Planck's Quantum Theory

10. The number of photons emitted in 10 hours by a 60 W sodium lamp (λ of photon = 6000 Å)

(A) 6.50×10^{24} (B) 6.40×10^{23}

(C) 8.40×10^{23} (D) 3.40×10^{23}

CAS374

11. The energy of a 700-nm photon is : -

(A) 1.77 eV (B) 2.47 Ev

(C) 700 eV (D) 3.57 eV

CAS375

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12. The energy required to break one mole of CI—CI bonds in $C1_2$ is 242 kJ mol^{-1}. The longest wavelength of light capable of breaking a single CI—CI bond is ($C = 3 \times 10^8$ ms^{-1} and $N_A = 6.02 \times 10^{23}$ mol^{-1})

(A) 494 nm (B) 594 nm (C) 640 nm (D) 700 nm

CAS376

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	B	A	D	D	C	B	D	A
Question	11	12								
Answer	A	A								

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Atomic Structure: RACE-02

Photoelectric Effect

1. A photon of energy hv is absorbed by a free electron of a metal having work - function $\phi < hv$. Choose the correct option(s).
(A) The electron is sure to come out.
(B) The electron is sure to come out with a kinetic energy $hv - \phi$
(C) Either the electron does not come out or it comes out with a kinetic energy $hv - \phi$
(D) It may come out with a kinetic energy less than and equal to $hv - \phi$

CAS377

2. Photoelectric emission is observed from a surface for frequencies ν_1 and ν_2 of the incident radiation ($\nu_1 > \nu_2$). If the maximum K.E. of the photoelectrons in two cases are in ratio 1 : K, then the threshold frequency ν_o is given by

(A) $\frac{\nu_2 - \nu_1}{K-1}$ (B) $\frac{K\nu_1 - \nu_2}{K-1}$ (C) $\frac{K\nu_2 - \nu_1}{K-1}$ (D) $\frac{\nu_2 - \nu_1}{K}$

CAS378

3. The work function of a metal is 4 eV. If EMR of 200 nm fall on the metal surface. Calculate the maximum K.E. of photo electron effect.

CAS379

4. Calculate the minimum & maximum kinetic energy in eV of photoelectrons produced in cesium by 400 nm light. The critical (maximum) wavelength for the photoelectric effect in cesium is 660 nm, when the potential difference is 2V.

CAS380

Bohr's Model for Hydrogen Atom

5. If the velocity of an electron in 1st orbit of H is V, then what will be the velocity in 4th orbit of Be^{+3} :-

(A) V (B) $\frac{V}{4}$ (C) 16 V (D) 4V

CAS381

6. Which of the following statement (regarding an atom of H) is correct -
(A) Kinetic energy of the electron is maximum in the first orbit
(B) Potential energy of the electron is maximum in the first orbit
(C) Radius of the second orbit is two times the radius of the first orbit
(D) Various energy levels are equally spaced

CAS382

7. Which is not a **CORRECT** order of energy for 1st 2nd & 3rd orbit :-
(A) $E_1 > E_2 > E_3$ (B) $(PE)_1 < (PE)_2 < (PE)_3$
(C) $(KE)_1 > (KE)_2 > (KE)_3$ (D) 'A' & 'C' both

CAS383

8. Match the following :-

(A) Energy of ground state of He^+	(i) +6.04 eV
(B) Potential energy of I orbit of H—atom	(ii) –27.2 eV
(C) Kinetic energy of II excited state of He	(iii) 8.72×10^{-18} J
(D) Ionisation potential of He	(iv) –54.4 eV

(A) A - (i), B - (ii), C - (iii), D - (iv) (B) A - (iv), B - (iii), C - (ii), D - (i)
(C) A- (iv), B - (ii), C - (i), D - (iii) (D) A - (ii), B - (iii), C - (i), D - (iv)

CAS384

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9. Choose the wrong one on the basis of Bohr's theory:-

(A) Velocity of electron $\propto \frac{1}{\sqrt{r}}$ (B) Frequency of revolution $\propto \frac{1}{n^3}$

(C) Velocity of electron $\propto \sqrt{r}$ (D) mvr $\propto$ n

CAS385

10. Which of the following statement is wrong for H-like atom -

(A) Magnitude of energy of an orbit in H-like species is directly proportional to l/n^2

(B) Frequency of revolution of an electron in an orbit is proportional to n^3

(C) Radius of a Bohr orbit is proportional to n^2

(D) Time period of revolution of an electron is proportional to n^3

CAS386

11. The orbital angular momentum of an electron is $\sqrt{3}\frac{h}{\pi}$. Which of the following may be the permissible value of angular momentum of this electron revolving in unknown Bohr orbit?

(A) $\frac{h}{\pi}$ (B) $\frac{h}{2\pi}$ (C) $\frac{3h}{2\pi}$ (D) $\frac{2h}{\pi}$

CAS387

12. A single electron orbits around a stationary nucleus of charge +Ze where Z is atomic number and 'e' is the magnitude of the electric charge. The hydrogen like species required 47.2 eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find

(i) the value of Z and give the hydrogen like species formed.

(ii) the kinetic energy (in eV) of the electron in the first Bohr orbit.

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CAS268

13. In the Bohr's model, for unielectronic species following symbols are used

$r_{n,z} \longrightarrow$ Radius of nth orbit with atomic number "z"

$U_{n,z} \longrightarrow$ Potential energy of electron in n^{th} orbit with atomic number "z"

$K_{n,z} \longrightarrow$ Kinetic energy of electron in n^{th} orbit with atomic number "z"

$v_{n,z} \longrightarrow$ Velocity of electron in n^{th} orbit with atomic number "z"

$T_{n,z} \longrightarrow$ Time period of revolution of electron in n^{th} orbit with atomic number "z"

Calculate z in all in cases.

(i) $U_{1,2} : K_{1,z} = -8 : 1$ (ii) $r_{1,z} : r_{2,1} = 1 : 8$

(iii) $v_{1,z} : v_{3,1} = 9 : 1$ (iv) $T_{1,2} : T_{2,z} = 9 : 32$

CAS272

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	D	B	2.2eV	2eV, 3.22 eV	A	A	A	C	C
Question	10	11	12		13				
Answer	B	D	(i) 5.00 ; (ii) 340.00		(i) 1, (ii) 2, (iii) 3, (iv) 3				

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Atomic Structure: RACE-03

Explanation of Line Spectrum of Hydrogen, Lyman, Balmer, Paschen Series

1. A dye absorbs a photon of wavelength λ and re-emits the same energy into two photons of wavelengths λ_1 and λ_2 respectively. The wavelength is related with and as :-
The energy of a 700-nm photon is :-

(A) $\lambda = \frac{\lambda_1 + \lambda_2}{\lambda_1 \lambda_2}$ (B) $\lambda = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ (B) $\lambda = \frac{\lambda_1^2 \lambda_2^2}{\lambda_1 + \lambda_2}$ (D) $\lambda = \frac{\lambda_1 \lambda_2}{(\lambda_1 + \lambda_2)^2}$

CAS388

2. A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, the other is at :-

(A) 743 nm (B) 518 nm (C) 1035 nm (D) 325

CAS389

3. The ratio of the energy of the electron in ground state of hydrogen to the electron in first excited state of Be^{3+} is :-

(A) 1 : 4 (B) 1 : 8 (C) 1 : 16 (D) 16 : 1

CAS390

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4. Ionisation energy of He^+ is 19.6×10^{-18} J $atom^{-1}$. The energy of the first stationary state (n = 1) of Li^{2+} is :-

(A) 8.82×10^{-17} J $atom^{-1}$ (B) 4.41×10^{-16} J $atom^{-1}$
(C) -4.41×10^{-17} J $atom^{-1}$ (D) -2.2×10^{-15} J $atom^{-1}$

CAS391

5. In hydrogen atom the energy difference between the states n = 2 and n = 3 is E eV . The ionisation potential of H atom is -

(A) 3.2 E (B) 5.6 E (C) 7.2 E (D) 13.2 E

CAS392

6. The LP. of an atom is 16 eV. What is the value of first excitation potential:-

(A) 10·2 eV (B) 12 eV (C) 14 eV (D) 16 eV

CAS393

7. In Balmer series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen ?

(A) $2 \rightarrow 5$ (B) $3 \rightarrow 2$ (C) $5 \rightarrow 2$ (D) $4 \rightarrow 1$

CAS394

8. The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be
(Rydberg constant= l.097× 107 m^{-1}) :

(A) 9.1×10^{-8} nm (B) 192 nm (C) 406 nm (D) 91 nm

CAS395

For More Join: @IITJEE_Advanced

9. Which transition emits photon of maximum frequency :-

(A) second spectral line of Balmer series (B) second spectral line of Paschen series

(C) fifth spectral line of Humphery series (D) first spectral line of Lymen series

CAS396

10. In an electronic transition H-atom cannot emit:-

(A) Visible light (B) γ- rays

(C) Infra red light (D) Ultra violet light

CAS397

11. Find the number of spectral lines in Paschen series emitted by atomic H, when electron is excited from ground state to 7th energy level returns back.

CAS398

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	A	C	C	B	C	D	D	B
Question	11									
Answer	4									

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Atomic Structure: RACE-04

Towards Quantum Mechanical Model of Atom; De Broglie's Hypothesis

1. The angular momentum J of the electron in a hydrogen atom is proportional to n^{th} power of (radius of the orbit) where n is :-

(A) 1 (B) –1

(C) $\frac{1}{2}$ (D) None

CAS399

2. The de Broglie wavelength of a tennis ball of mass 60 g moving with a velocity of 10 metres per second is approximately :

(A) 10^{-25} metres (B) 10^{-33} metres

(C) 10^{-31} metres (D) 10^{-16} metres

CAS400

3. A particle moving with a velocity of 6.626×10^7 m/sec has a De-Broglie wavelength of 1 A in a circular path of radius 0.529 A. The angular momentum of particle is.

($h = 6.626 \times 10^{-34}$ J × sec).

(A) 3.5×10^{-34} kg m^2sec^{-1} (B) 3.5×10^{-33} kg m^2sec^{-1}

(C) 3.5×10^{-31} kg m^2sec^{-1} (D) 1.053×10^{-35} kg m^2sec^{-1}

CAS401

Heisenberg's Uncertainty Principle

4. An atom has a mass of 0.02 kg and uncertainty in its velocity is 9.218×10^{-6} m/sec then uncertainty in position is ($h = 6.626 \times 10^{-34}$ Js)

(A) 2.86×10^{-23} m (B) 2.86×10^{-32}cm

(C) 1.5×10^{-27} m (D) 3.9×10^{-10}m

CAS402

5. Uncertainty in position of a particle of 25 g in space is 10^{-5} m. Hence uncertainty in velocity (ms^{-1}) is (Planck's constant $h = 6.6 \times 10^{-34}$ Js)

(A) 2.1×10^{-23} (B) 2.1×10^{-34}

(C) 0.5×10^{-34} (D) 5.0×20^{-24}

CAS403

6. Uncertainty in the position of an electron (mass = 9.1×10^{-31} Kg) moving with a velocity 300 ms^{-1}, accurate upto 0.001 %, will be:- ($h = 6.63 \times 10^{-34}$ Js)

(A) 5.76×10^{-2} m (B) 1.92×10^{-2} m

(C) 3.84×10^{-2} m (D) 19.2×10^{-2} m

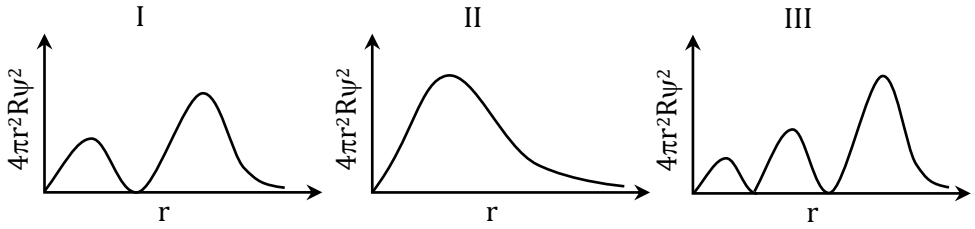
CAS404

Bankers Academy JEE

For More Join: @IITJEE_Advanced

Quantum Mechanical Model of Atom (Schrodinger wave equation using Polar coordinates; Graph transformation; Graphs of Schrodinger wave equation; Equations of different orbitals)

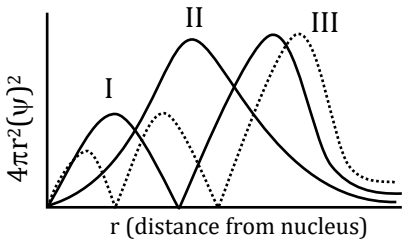
7. Which of following has correct matching of curve and orbital



(A) I(2p) II(1s) III(4p)
(B) I(3p) II(3d) III(3s)
(C) I(4d) II(2p) III(5d)
(D) I(2s) II(4f) III(3d)

CAS405

8. Consider the following radial distribution function diagrams. Which of the following has the correct matching of curve and orbital ?



(A) I(3s), II(3p), III(3d)
(B) I(3d), II(3p), III(3s)
(C) I(3p), II(3d), III(3s)
(D) I(3s), II(3d), III(3p)

Hinkers Academy JEE

CAS406

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	C	B	A	A	A	B	BC	C

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Atomic Structure: RACE-05

Quantum numbers

1. Principal, azimuthal and magnetic quantum numbers are respectively related to
(A) size, shape and orientation (B) shape, size and orientation
(C) size, orientation and shape (D) none of these

CAS407

2. How many maximum electrons can be described by the quantum numbers n = 5, ℓ = 2 in a particular atom?
(A) 2 (B) 6 (C) 10 (D) 14

CAS408

3. If an electron has spin quantum number of + $\frac{1}{2}$ and magnetic quantum number of –1 it cannot be present in -
(A) f-orbital (B) d-orbital (C) p-orbital (D) s-orbital

CAS409

4. When the quantum number n,*l*,m,m_s are represented by 3,3,2,+1/2, the symbolism for the electron is -
(A) 3s (B) 3d
(C) 3f (D) impossible set of quantum number

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CAS410

5. For a 6s electron the values of n,*l*,m,m_s respectively could be:
(A) 6,4,4,+1/2 (B) 1,0,0,+1/2 (C) 6,1,0,+1/2 (D) 6,0,0, +1/2

CAS411

6. Which one of the following sets of quantum numbers (n,*l*,m,m_s) represents an impossible arrangement?
(A) 3,2,–2,+1/2 (B) 4,0,0,+1/2 (C) 3,2,–3,+1/2 (D) 5,3,0,–1/2

CAS412

7. What type of orbital is designated n = 2, ℓ = 3, m_ℓ = –2 ?
(A) 4p (B) 4d (C) 4f (D) None

CAS413

8. ('*l*' and '*m*' are respectively the azimuthal and magnetic quantum numbers)

Column-I		Column-II	
(A)	Total number of values of (*l*) for a shell	(P)	0, 1, 2, ...(n–1)
(B)	Values of (*l*) for a shell	(Q)	+*l*, ...+2, +1, 0, –1, –2, ...–*l*
(C)	Total number of values of (*m*) for a subshell	(R)	(2*l* + 1)
(D)	Values of (*m*) for a subshell	(S)	n

CAS414

For More Join: @IITJEE_Advanced

9. Which of the following statements regarding subshell filling order for a neutral atom is/are correct ?
(I) Electrons are assigned to the 4s subshell before they are assigned to the 3d subshell
(II) Electrons are assigned to the 4f subshell before they are assigned to the 6s subshell
(III) Electrons are assigned to the 4d subshell before they are assigned to the 5p subshell
(A) I only (B) II only (C) I and III (D) I, II and III

CAS415

10. **Statement-1 :** For n = 2 the values of ℓ may be 0, 1 and m may be 0, ±1.

Statement-2 : For each value of n, there are 0 to (n – 1) possible values of ℓ, for each value of ℓ there are 0 to ± ℓ values of m.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CAS416

11. Supposing the I.P. of hydrogen atom is 960 eV. Find out the value of principal quantum number having the energy equal to – 60 eV :
(A) n = 2 (B) n = 3 (C) n = 4 (D) n = 5

CAS417

12. The orbital angular momentum for an electron revolving in an orbit is given by . $\sqrt{\ell(\ell+1)} \cdot \frac{h}{2\pi}$. This momentum for an s-electron will be given by

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(A) $\sqrt{2} \cdot \frac{h}{2\pi}$ (B) $+\frac{1}{2} \cdot \frac{h}{2\pi}$ (C) zero (D) $\frac{h}{2\pi}$

CAS418

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	A	C	D	D	D	C	D	(A) → S; (B) → P; (C) → R ; (D) → Q
Question	9	10	11	12				
Answer	C	A	C	C				

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Atomic Structure: RACE-06

Rules for Electronic Configuration; Aufbau's Principle; Pauli's Exclusion Principle; Hund's Rule; Half and fully filled subshell configuration; All exceptional configuration; Degeneracy

1. Degenerate atomic orbitals have

(A) equal energy (B) nearly equal energy

(C) different energy (D) none of the above

CAS419

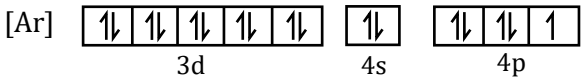
2. Any p-orbital can accomodate up to

(A) four electrons (B) Two electrons in parallel spin

(C) Six electrons (D) Two electrons with opposite spin

CAS420

3. What is a possible set of quantum numbers for the unpaired electron in the orbital box diagram below ?



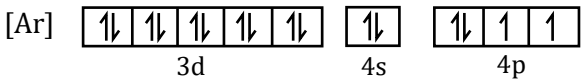
(A) $n = 1, \ell = 1, m_\ell = -1, m_s = +1/2$ (B) $n = 4, \ell = 1, m_\ell = -1, m_s = +1/2$

(C) $n = 4, \ell = 2, m_\ell = -2, m_s = +1/2$ (D) $n = 4, \ell = 0, m_\ell = 0, m_s = +1/2$

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CAS421

4. Which element has the following ground state electron configuration ?



(A) Se (B) As (C) S (D) Ge

CAS422

5. Hund's rule states that the most stable arrangement of electrons (for a ground state electron configuration)

(A) has three electrons per orbital, each with identical spins

(B) has m_ℓ values greater than or equal to +1

(C) has the maximum number of unpaired electrons, all with the same spin in degenerate orbital

(D) has two electrons per orbital, each with opposing spins

CAS423

6. The Pauli's exclusion principle states that

(A) no two electrons in an atom can have the same four quantum numbers

(B) electrons can have either ±1/2 spins

(C) electrons with opposing spins are attracted to each other

(D) None of these

CAS424

For More Join: @IITJEE_Advanced

7. Choose the correct option for the quantum numbers of the last electron of $3p^6$.

(A) 4, 0, 0, +1/2 (B) 3, 1, –1, –1/2 (C) 4, 1, 0, $-\frac{1}{2}$ (D) 3, 0, 1, $\frac{1}{2}$

CAS425

8. Consider the ground state of Cr (Z = 24). The numbers of electrons with the azimuthal quantum numbers l = 1 and 2 respectively are

(A) 16 and 4 (B) 12 and 5 (C) 12 and 4 (D) 16 and 5

CAS426

9. The total number of electrons in Cr atom for which m = 0

(A) 1 (B) 8 (C) 12 (D) 16

CAS427

10. How many maximum possible set(s) of quantum no. are possible for 6th electron of Fe

(A) 1 (B) 3 (C) 6 (D) 10

CAS428

11. The maximum no. of electron in phosphorous atom for which n + l + m = 3 will be -

(A) 6 (B) 5 (C) 4 (D) 3

CAS429

12. H-atom have infinite shells, write total number of shells which does not contain f-subshell.

CAS430

13. In multielectronic atom, maximum number of degenerated orbitals present in 3rd shell

CAS431

14.

Column-I (Element)		**Column-II (Max. value of n and λ respectively ; Consider filled subshell only)**	
P	P	1	6, 2
Q	N	2	3, 1
R	Pb	3	6, 3
S	Cs	4	2, 1

Code :

P	**Q**	**R**	**S**
(A) 4	1	2	3
(B) 2	4	1	3
(C) 3	1	2	4
(D) 2	4	3	1

CAS432

15. Which of the following having maximum number of unpaired electron -

(A) Na^+ (B) N^{3-} (C) Fe^{3+} (D) Cr^{3+}

CAS433

ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Answer	A	D	B	A	C	A	B	B	C	C
Question	**11**	**12**	13	14	15					
Answer	B	3	5	D	C					

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